

# Ethan Lu

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## EDUCATION

### Georgia Institute of Technology

Bachelor of Science in Computer Science

Atlanta, GA

Expected Dec 2026

- **GPA:** 4.00 | **Concentrations:** Information Internetworks and Intelligence
- **Relevant Coursework:** Data Structures and Advanced Algorithms, Systems and Networks, Computer Architecture, Database Systems, HCI Research Methods, Discrete Mathematics, Linear Algebra, Java OOP

## PROJECTS

### factr.ai

Founder

Atlanta, GA

May 2025 – Present

- Reduced misinformation detection false negatives by ~20% over unimodal baselines by developing a multimodal AI pipeline cross-validating BERT, CLIP, and AudioSet models
- Deployed a scalable FastAPI service on Azure, integrating Docker for modular experimentation and preparing for future public API access
- Trained models on LIAR, DFDC, and other benchmark datasets to ensure cross-modal robustness for real-world misinformation scenarios

### The Climate Reality Project

Web Co-Developer

Sparta, NJ

Mar 2022 – Oct 2022

- Expanded access to 100+ climate education resources integrated into the curricula of 15 local schools and libraries upon live, in-person demonstrations, by developing a searchable HTML/CSS/JavaScript website
- Enabled real-time content publishing by integrating the TinyMCE rich text editor, allowing multiple teammates to upload and update educational material directly to the site

### Senior Home Sign-In Application

Full-Stack Co-Developer

Springfield, NJ

Dec 2021 – Dec 2022

- Improved accuracy and efficiency of daily resident and visitor log management at a local senior community, by developing a Python-based application with an accessible Tkinter interface tailored for elderly users
- Implemented JSON-based data storage to enable efficient read/write operations and offline admin access

## EXPERIENCE

### Data-Driven Education, Vertically Integrated Project

Lead ML/Data Engineer

Atlanta, GA

Jan 2025 – Present

- Increased scalability of model training by generating a synthetic dataset with ~90% statistical similarity to the institution's course forum data, using OpenAI's o1 model and validating similarity through Latent-Dirichlet Allocation (LDA) topic modeling
- Improved instructor insight by building a React dashboard with pyLDAvis visualizations for topic modeling, with planned integration of sentiment analysis
- Developing a BERT-based classifier to categorize forum posts into cognitive presence categories, increasing accuracy and interpretability of discussion analytics

### MathWorks Math Modeling Challenge

Lead Data Analyst and Scientist

Philadelphia, PA

Mar 2024

- Achieved a predictive accuracy of 0.86 AUC in forecasting homelessness trends in U.S. cities by developing a binomial logistic growth model to analyze complex socioeconomic datasets using Pandas and MATLAB.

## LEADERSHIP/COMMUNITY ENGAGEMENT

### Investment Management Club

Tech+Commodity Sector Researcher and Analyst

Atlanta, GA

Aug 2024 – Present

- Delivered interpretable 0–1 scaled growth potential and market impact scores for companies in a simulated \$100K investment portfolio, by designing a scoring framework using Gaussian Mixture Models and log-probability transformations
- Built data pipelines to collect and normalize real-time and historical financial metrics from the Financial-ModelingPrep API, enabling unsupervised sector-based performance analysis with Python and PyTorch

### Data Science @ Georgia Tech

Sports Analysis Project Data Scientist

Atlanta, GA

Aug 2024 – Dec 2024

- Determined that investing in fewer, high-impact rather than many, low-impact players correlated with higher league survival rates, based on analysis of ~500 Premier League transfer records (2008–2024) filtered from a dataset of 75K+ entries using BeautifulSoup and visualizing results in Seaborn and Matplotlib

## PROFESSIONAL SKILLS

- **Languages:** Python, Java, JavaScript, SQL, C, MATLAB, HTML
- **Frameworks & Libraries:** PyTorch, React, Pandas, NumPy, MySQL
- **Data Analysis & Visualization:** BeautifulSoup, Matplotlib, Seaborn, pyLDAvis, JupyterNB, Tkinter
- **AI & Tools:** BERT, CLIP, Hugging Face, FastAPI, Docker, VSCode, IntelliJ, CLion, Jira, Notion